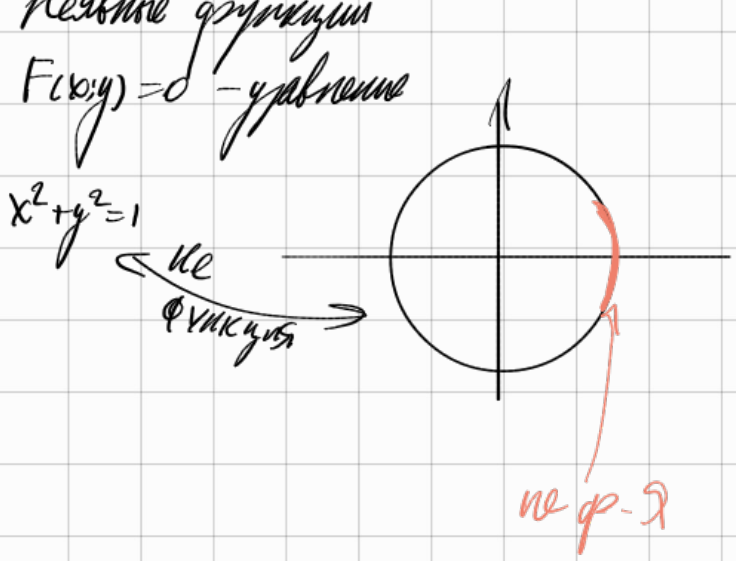
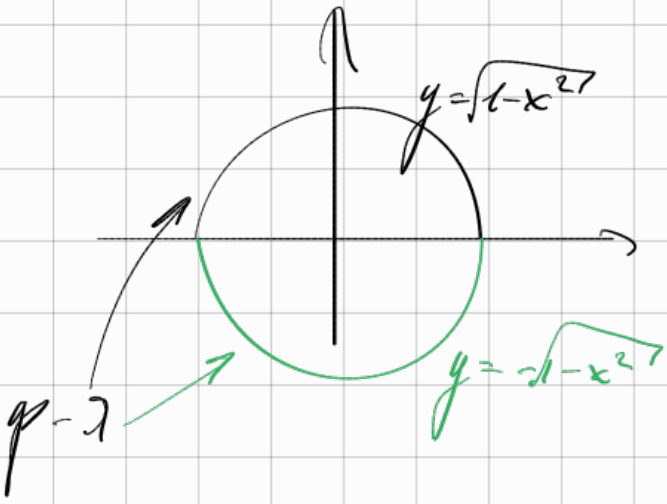
Неявные функции

$F(x;y)=0$ – уравнение

$x^2+y^2=1$





Теорема о неявной ф-ии

Если $F(x;y)$ непр в $U\delta_0(x_0;y_0)$ $F(x_0;y_0)=0$ и F'_y непр в $(x_0;y_0)$ и $F'_y(x_0;y_0)\neq 0$, тогда $\exists\, U\delta_1(x_0)$ и $U\delta_2(y_0)$ $\forall x\in U\delta_1(x_0)$ $\exists! y\in U\delta_2(y_0)$ ↳

↳ $F(x;y)=0$

Задана $y=f(x)$

Следствие

Если в дополнение F'_x $\exists$ в ок-ти $U\delta_0(x_0;y_0)$ и непр-на в $(x_0;y_0)$, то f дир в x_0, причем

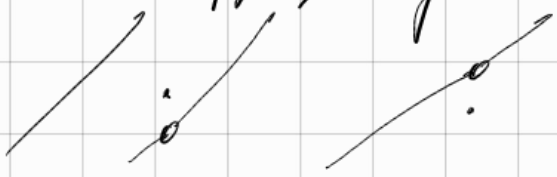
$$f'(x_0)=-\frac{F'_x(x_0;y_0)}{F'_y(x_0;y_0)}$$

Если в дополнение F'_x и F'_y непр в $U\delta(x_0;y_0)$, то f в нек. ок. имеет непр производную:

$$f'(x)=-\frac{F'_x(x;f(x))}{F'_y(x;f(x))}$$

Т1 $x^2=y^2$

а) Сколько функций $y:\mathbb{R}\to\mathbb{R}$ удовл уравн?



более чем счетное

б) Сколько непрерывных ф-ий $y:\mathbb{R}\to\mathbb{R}$ удовл уравн?

4



в) сколько нет ф-ий $y: \mathbb{R} \to \mathbb{R}$ удов. усл. $y(1)=1$? ✓

2) сколько нет ф-ий $y: [1;2] \to \mathbb{R}$ удов. усл. $y(1)=1$? 2

1 ✓

Теорема

$F(x;y): U\delta_0(x^{(0)};y^{(0)}) \to \mathbb{R}$ $\quad x^{(0)} \in \mathbb{R}^n$, $y^{(0)} \in \mathbb{R}$

Если $F(x;y)$ непр. диф. на $U\delta_0(x^{(0)};y^{(0)})$, $F(x^{(0)};y^{(0)})=0$

и $F'_y(x^{(0)};y^{(0)}) \neq 0$, то $\exists\, U\delta_1(x^{(0)})$ $\exists\, U\delta_2(y^{(0)})$ $\forall x \in U\delta_1(x^{(0)})$ $\exists!\, y \in U\delta_2(y^{(0)}) \hookrightarrow$

$\hookrightarrow F(x;y)=0$ т.е. на $U\delta_1(x^{(0)})$ задана $y=f(x)$

При этом f непр. диф. в $U\delta_1(x^{(0)})$ и $f'_{x_i}(x) = -\dfrac{F'_{x_i}(x;f(x))}{F'_y(x;f(x))}$

§ 3 №61(г) $\quad F(x;y;u)=0$

$x^2 - 2y^2 + 3u^2 - yu + y = 0$

а) $\left(1;1;\frac{1}{3}\right)$ $\quad u(x;y)$

$F'_u = 6u - y$

$F'_u(1;1;\frac{1}{3}) = 1 \neq 0$

$F'_x = 2x \qquad F'_y = -u - u + 1$

$F'_x(1;1;\frac{1}{3}) = 2 \qquad F'_y(1;1;\frac{1}{3}) = -u - \frac{1}{3} + 1 = -\frac{10}{3}$

$u'_x = -\dfrac{F'_x}{F'_u} = -\dfrac{2}{1} = -2$

$u'_y = -\dfrac{F'_y}{F'_u} = -\dfrac{10}{3 \cdot 1} = -\dfrac{10}{3}$

§3 №61(г)

$u^3 - xu + y = 0$

а) $(3;-2;2)$ $\quad u'_x = -\dfrac{F'_x}{F'_u} = \dfrac{u}{3u^2 - x} = \dfrac{2}{12-3} = \dfrac{2}{9}$

$u'_y = -\dfrac{F'_y}{F'_u} = \dfrac{-1}{3u^2 - x} = -\dfrac{1}{9}$ $\quad du = \dfrac{2}{9}dx - \dfrac{1}{9}dy$

Система уравнений

$$F_1(x_1 \dots x_n\ y_1 \dots y_m) = 0$$
$$\vdots$$
$$F_m(x_1 \dots x_n\ y_1 \dots y_m) = 0$$

$\vec{x} = (x_1 \dots x_n)^T$

$\vec{y} = (y_1 \dots y_m)^T$

$\vec{F} = (F_1 \dots F_m)^T$

$\vec{F}(\vec{x}; \vec{y}) = \vec{0}$

$$\frac{\partial \vec{F}}{\partial \vec{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \dots & \frac{\partial F_1}{\partial y_m} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial y_1} & \dots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}$$

$$\frac{\partial \vec{F}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}$$

Теорема

$\exists \vec{F}(\vec{x}; \vec{y})$ нелин диф в $U_{\delta_0}(\vec{x}^{(0)}; \vec{y}^{(0)})$, причем $\vec{F}(\vec{x}^{(0)}; \vec{y}^{(0)}) = \vec{0}$ и

$\det \frac{\partial \vec{F}}{\partial \vec{y}}(\vec{x}^{(0)}; \vec{y}^{(0)}) \neq 0$

тогда $\exists U_{\delta_1}(\vec{x}^{(0)})$ и $\exists U_{\delta_2}(\vec{y}^{(0)})$: $\forall \vec{x} \in U_{\delta_1}(\vec{x}^{(0)})$ $\exists (!)\ \vec{y} \in U_{\delta_2}(\vec{y}^{(0)}) \hookrightarrow \vec{F}(\vec{x}; \vec{y}) = 0$,

т.е. на $U_{\delta_1}(\vec{x}^{(0)})$: $\vec{y} = \vec{f}(\vec{x})$, причем $\vec{f}$ - нелин диф в $\frac{\partial \vec{f}}{\partial \vec{x}} = -\left[\frac{\partial \vec{F}}{\partial \vec{y}}\right]^{-1}\left[\frac{\partial \vec{F}}{\partial \vec{x}}\right]$

§3 №25

(1;2) $u(x;y)$ $v(x;y)$

$$x e^{u+v} + 2uv = 1$$
$$y e^{u-v} - \frac{u}{1+v} = 2x$$
$$u(1;2) = v(1;2) = 0$$

$$F_1(x\ y\ u\ v) = x e^{u+v} + 2uv - 1 = 0$$
$$F_2(x\ y\ u\ v) = y e^{u-v} - \frac{u}{1+v} - 2x = 0$$

$$\frac{\partial \vec{F}}{\partial \vec{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix}\Bigg|_{\dots} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \quad \det = -3$$

$$\frac{\partial F_1}{\partial u} = x e^{u+v} + 2v = 1 \qquad \frac{\partial F_1}{\partial v} = x e^{u+v} + 2u = 1$$

$$\frac{\partial F_2}{\partial u} = y e^{u-v} - \frac{1}{1+v} = 1 \qquad \frac{\partial F_2}{\partial v} = -y e^{u-v} + \frac{u}{(1+v)^2} = -2$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\Delta}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \qquad \left(\frac{\partial \vec{F}}{\partial \vec{y}}\right)^{-1} = \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\left(\frac{\partial \vec{F}}{\partial \vec{x}}\right)\Big|_{\dots} = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix}\Big|_{\dots} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

$$\frac{\partial F_1}{\partial x} = e^{u+v} = 1 \qquad \frac{\partial F_1}{\partial y} = 0 \qquad \frac{\partial F_2}{\partial x} = -2 \qquad \frac{\partial F_2}{\partial y} = e^{u-v} = 1$$

$$\vec{f} = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u(x;y) \\ v(x;y) \end{pmatrix}$$

$$\frac{\partial \vec{f}}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\frac{\partial u}{\partial x} = 0 \qquad \frac{\partial u}{\partial y} = -\frac{1}{3} \qquad \frac{\partial v}{\partial x} = -1 \qquad \frac{\partial v}{\partial y} = \frac{1}{3}$$

12.09

$$F(x;y;u) = 0$$

$$u'_x = -\frac{F'_x(x;y;u)}{F'_u(x;y;u)} \qquad u'_y = -\frac{F'_y(x;y;u)}{F'_u(x;y;u)}$$

$$u''_{xx} = (u'_x)'_x = -\left(\frac{F'_x(x;y;u(x;y))}{F'_u(x;y;u(x;y))}\right)'_x =$$

$$= -\frac{(F''_{xx}(x;y;u(x;y)) + F''_{xu}(x;y;u(x;y))\,u'_x(x;y))\,F'_u(x;y;u(x;y)) - F''_{ux}(x;y;u(x;y)) +}{(F'_u(x;y;u(x;y)))^2}$$

$$+ \frac{F''_{uu}(x;y;u(x;y))\,u'_x(x;y)\,F'_x(x;y;u(x;y))}{(F'_u(x;y;u(x;y)))^2}$$

§4 №3(5)

Найти d^2u $\qquad (x;y;u) = (1;-1;-1)$

$u(1;-1) = -1$

$F(x;y;u) = u^3 + 2yu + xy = 0$

$F'_x = y \qquad F'_y = 2u + x \qquad F'_u = 3u^2 + 2y$

$F'_x|_{(1;-1;-1)} = -1 \qquad F'_y|_{(1;-1;-1)} = 1 \qquad F'_u|_{(1;-1;-1)} = 1$

$u'_x = -\frac{F'_x}{F'_u} = 1$

$u'_y = -\frac{F'_y}{F'_u} = 1$

$du|_{(1;-1)} = dx + dy$

$u''_{xx} = \frac{y}{(3u^2(x;y) + 2y)^2} \quad (6u(xy)\,u'_x(x;y))$

$u''_{xx}|_{(1;-1)} = 6$

$u''_{xy} = (u'_x)'_y = \frac{-(3u^2(x;y) + 2y) - 6u(x;y)u'_y(x;y) + 2y}{(3u^2(x;y) + 2y)^2}\Big|_{(1;-1)} = 3$

$u''_{yy} = \ldots 2$

$d^2u|_{(1;-1)} = 6dx^2 + 6dxy + 2dy^2$

§4 №46(1)

$f(x+u;y+u) = 0 \quad f \in C^2;\ u \in C^2$

$F(x;y;u) = f(x+u;y+u) = 0$

$f = f(a;b) \qquad \frac{\partial f}{\partial a} = f'_1 \qquad \frac{\partial f}{\partial b} = f'_2$

$F'_x = f'_1 \qquad F'_y = f'_2 \qquad F'_u = f'_1 + f'_2$

$u'_x = -\frac{F'_x}{F'_u} = -\frac{f'_1}{f'_1 + f'_2} \qquad u'_y = -\frac{f'_2}{f'_1 + f'_2}$

$u'_x = -\frac{f'_1(x+u(x;y);y+u(x;y))}{f'_1(x+u(x;y);y+u(x;y)) + f'_2(x+u(x;y);y+u(x;y))}$

$du = -\frac{f'_1}{f'_1 + f'_2}dx - \frac{f'_2}{f'_1 + f'_2}dy$

$u-p-\lambda(x;y)$

$u''_{xx} = -\frac{(f''_{11}\cdot[1+u'_x] + f''_{12}u'_x)(f'_1 + f'_2) - (f''_{11}[1+u'_x] + f''_{12}u'_x + f''_{21}[1+u'_x] + f''_{22}u'_x)f'_1}{(f'_1 + f'_2)^2}$

$$u''_{xy} = -\frac{(f''_{11}u'_y + f''_{12}[1+u'_y])(f'_1+f'_2) - (f''_{11}u'_y + f''_{12}[1+u'_y] + f''_{21}u'_y + f''_{22}[1+u'_y])f'_1}{(f'_1+f'_2)^2}$$

$$u''_{yy} = -\frac{(f''_{21}u'_y + f''_{22}[1+u'_y])(f'_1+f'_2) - f''_{11}u'_y + f''_{12}[1+u'_y] + f''_{21}u'_y + f''_{22}[1+u'_y])f'_2}{(f'_1+f'_2)^2}$$

Дифференцируемое отображение

Опр $\exists f$ лин. нулевое, ставящее каждому $x \in G \subset \mathbb{R}^n$ в соотв $y \in \mathbb{R}^m$
тогда f - отображение, обозн $f: G \subset \mathbb{R}^n \to \mathbb{R}^m$

$$f \Leftrightarrow \begin{cases} y_1 = f_1(x_1; \dots; x_n) \\ \vdots \\ y_m = f_m(x_1; \dots; x_n) \end{cases}$$

$f_i(x_1; \dots; x_n)$ координатные функции

Опр (Диф) $f: U_\delta(x_0) \subset \mathbb{R}^n$

$x_0 = (x_0^1; \dots; x_0^n)$

Говорят, что f диф в т. x_0, если $f(x) - f(x_0) = A \cdot \Delta x + o(|\Delta x|)$, $\Delta x \to 0$ $\quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

A - матрица $m \times n$

$$\Delta x = \begin{pmatrix} x^1 - x_0^1 \\ \vdots \\ x^n - x_0^n \end{pmatrix}$$

$$|\Delta x| = \sqrt{\sum_{i=1}^{n} (\Delta x^i)^2}$$

Замечание Если f диф в x_0 $\quad A = \begin{pmatrix} \frac{\partial f_1}{\partial x^1} & \dots & \frac{\partial f_1}{\partial x^n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x^1} & \dots & \frac{\partial f_m}{\partial x^n} \end{pmatrix}$ - матрица Якоби

Если $m = n$, то A - квадр.
$\det A$ - якобиан

Теорема $\exists f: U_\delta(x^0) \subset \mathbb{R}^n \to \mathbb{R}^n$ $\quad J$ матрица Якоби
$\exists f$ - непр диф в этой окрестности, $\det J|_{x_0} \neq 0$
Тогда $\exists U_\varepsilon(y_0), y_0 = f(x_0)$ определено обратное отображение, причем оно непр диф

Т3 $f: \mathbb{R}^2 \to \mathbb{R}^2$

$u = e^x \cos y$

$v = e^x \sin y$

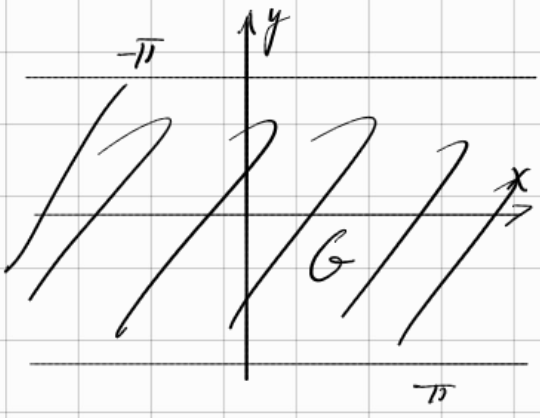
$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$$

$\det J = e^{2x}\cos^2 y + e^{2x}\sin^2 y = e^{2x} \neq 0$

$\begin{pmatrix} u \\ v \end{pmatrix}(x;y) = \begin{pmatrix} u \\ v \end{pmatrix}(x; y+2\pi)$

не взаимно однозначно на $\mathbb{R}^2$

Ч-во значений отображения $\mathbb{R}^2 \setminus (0;0)$



Обл $G = \mathbb{R}^2 \setminus (0;0)$

$\mathbb{R}^2 \setminus (0;0) \xleftrightarrow[\text{однозн}]{\text{вз}} (-\infty; +\infty) \times [-\pi; \pi]$

Т4

$F: \mathbb{R}^2 \to \mathbb{R}^2$

$x = r\cos\varphi$

$y = r\sin\varphi \quad : r > 0$

$\Leftrightarrow \varphi = \varphi(x;y)$

$r = r(x;y)$

$$J = \begin{pmatrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{pmatrix}$$

$r = \sqrt{x^2+y^2}$

$\operatorname{tg}\varphi = \frac{y}{x}$

$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{r}$ $\quad \frac{\partial r}{\partial y} = \frac{y}{r}$

$\operatorname{tg}\varphi = \frac{y}{x}$ $\quad \frac{1}{\cos^2\varphi}\cdot\varphi' = -\frac{y}{x^2}$ $\quad \varphi' = -\frac{y\cos^2\varphi}{x^2} = -\frac{y}{r^2}$

$\frac{1}{\cos^2\varphi}\cdot\varphi' = \frac{1}{x}$ $\quad \varphi' = \frac{\cos^2\varphi}{x} = \frac{x}{r^2}$

$$J = \begin{pmatrix} r'_x & r'_y \\ \varphi'_x & \varphi'_y \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$$

Другой способ

$y = f(x) \quad x = g(y) \quad J_f = J_g^{-1}$

$x = x(r;\varphi)$
$y = y(r;\varphi)$

$x'_r = \cos\varphi \quad x'_\varphi = -r\sin\varphi$
$y'_r = \sin\varphi \quad y'_\varphi = r\cos\varphi$

$$J_{(r;\varphi)\to(x;y)} = \begin{pmatrix} x'_r & x'_\varphi \\ y'_r & y'_\varphi \end{pmatrix} = \begin{pmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{pmatrix} \qquad \det J_{(r;\varphi)\to(x;y)} = r$$

$$J_{(x;y)\to(r;\varphi)} = J^{-1}_{(r;\varphi)\to(x;y)} = \frac{1}{\det}\begin{pmatrix} r\cos\varphi & r\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} = \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\frac{\sin\varphi}{r} & \frac{\cos\varphi}{r} \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r} \end{pmatrix}$$

19.09

§3 № 86

$$x\frac{\partial u}{\partial y} - y\frac{\partial u}{\partial x} = 0$$

$x = r\cos\varphi$
$y = r\sin\varphi$

$$\frac{\partial r}{\partial x} = \frac{x}{r} \qquad \frac{\partial \varphi}{\partial x} = -\frac{y}{r^2}$$
$$\frac{\partial r}{\partial y} = \frac{y}{r} \qquad \frac{\partial \varphi}{\partial y} = \frac{x}{r^2}$$

$u = u(x;y) = \tilde{u}(r;\varphi) = \tilde{u}(r(x;y);\varphi(x;y))$

$$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

Подставляем

$$x\left(\frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right) - y\left(\frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right) = \cancel{\frac{xy}{r}\frac{\partial \tilde{u}}{\partial r}} + \frac{x^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi} - \cancel{\frac{xy}{r}\frac{\partial \tilde{u}}{\partial r}} + \frac{y^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi} =$$

$$= \frac{\partial \tilde{u}}{\partial \varphi} = 0$$

$\tilde{u} = f(r) \quad f \in C^1$ (непр. диф.)

$u = f(\sqrt{x^2+y^2}), \quad f \in C$

§3 №30

Преобразовать $(y-z)\frac{\partial z}{\partial x}+(y+z)\frac{\partial z}{\partial y}=0$ считая x за ф-цию,

а $\begin{cases} u=y-z \\ v=y+z \end{cases}$ за идут

б) $y=\frac{v+u}{2}$

$z=\frac{v-u}{2}$

$dy=\frac{1}{2}du+\frac{1}{2}dv$

$dz=-\frac{1}{2}du+\frac{1}{2}dv$

2) $x=x(u;v)$ – хотим

$dx=\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv$

3) $z=z(x;y)$ – есть

$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$

Доказано Шмегерман

4) 2) → 3)

$dz=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv\right)+\frac{\partial z}{\partial y}\left(\frac{1}{2}du+\frac{1}{2}dv\right)$

$dz=\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{1}{2}\frac{\partial z}{\partial y}\right)du+\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{1}{2}\frac{\partial z}{\partial y}\right)dv$

5) сравнивая 4 и 2

$\frac{\partial z}{\partial x}\frac{\partial x}{\partial y}+\frac{1}{2}\frac{\partial z}{\partial y}=-\frac{1}{2}$

$\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{1}{2}\frac{\partial z}{\partial y}=\frac{1}{2}$

$\frac{\partial z}{\partial x}=\frac{1}{\frac{\partial x}{\partial v}-\frac{\partial x}{\partial u}}$ $\quad \frac{\partial z}{\partial y}=\frac{\frac{\partial x}{\partial u}+\frac{\partial x}{\partial v}}{\frac{\partial x}{\partial u}-\frac{\partial x}{\partial v}}$

6) 5 в уравнение

$u\left(\frac{\partial x}{\partial v}-\frac{\partial x}{\partial u}\right)^{-1}+v\left(\frac{\partial x}{\partial v}-\frac{\partial x}{\partial u}\right)^{-1}\left(\frac{\partial x}{\partial u}+\frac{\partial x}{\partial v}\right)=0$

$v\frac{\partial x}{\partial u}+v\frac{\partial x}{\partial v}=u$

№51(2)

$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=0$

$\begin{cases} x=r\cos\varphi \\ y=r\sin\varphi \end{cases}$ $\quad u=u(x;y)=\tilde{u}(r;\varphi)=\tilde{u}(r(x;y);\varphi(x;y))$

$\frac{\partial u}{\partial x}=\frac{\partial\tilde{u}}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial\tilde{u}}{\partial\varphi}\frac{\partial\varphi}{\partial x}=\frac{x}{r}\frac{\partial\tilde{u}}{\partial r}-\frac{y}{r^2}\frac{\partial\tilde{u}}{\partial\varphi}$

$\frac{\partial^2 u}{\partial x^2}=\left(\frac{1}{r}-\frac{x^2}{r^3}\right)\frac{\partial\tilde{u}}{\partial r}+\frac{x}{r}\left(\frac{\partial^2\tilde{u}}{\partial r^2}\frac{x}{r}-\frac{\partial^2\tilde{u}}{\partial r\partial\varphi}\frac{y}{r^2}\right)+\frac{2yx}{r^4}\frac{\partial\tilde{u}}{\partial\varphi}-\frac{y}{r^2}\left(\frac{\partial^2\tilde{u}}{\partial r\partial\varphi}\frac{x}{r}-\frac{\partial^2\tilde{u}}{\partial\varphi^2}\frac{y}{r^2}\right)$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{-xy}{r^3}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\cdot\frac{x}{r^2} - \left(\frac{1}{r^2} - \frac{2y^2}{r^4}\right)\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial \varphi \partial r}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right)\right.$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{r} - \frac{y^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{y}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{x}{r^2}\right) - \frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right)$$

$$x^2\left(\left(\frac{r}{r} - \frac{x^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{y}{r^2}\right) + \frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{y}{r^2}\right)\right) + 2xy\left(-\frac{xy}{r^3}\frac{\partial \tilde{u}}{\partial r} + \right.$$

$$+ \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{x}{r^2}\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\right) - \left(\frac{1}{r^2} - \frac{2y^2}{r^4}\right)\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{y}{r}\frac{\partial^2 \tilde{u}}{\partial \varphi \partial r} + \frac{x}{r^2}\frac{\partial^2 \tilde{u}}{\partial \varphi^2}\right) +$$

$$+ y^2\left(\left(\frac{1}{r} - \frac{y^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{y}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{\partial^2 u}{\partial r \partial \varphi}\frac{x}{r^2}\right) - \frac{2xy}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right)\right) = 0$$

$$\frac{\partial \tilde{u}}{\partial r}\left(\frac{x^2y^2}{r^3} - \frac{2x^2y^2}{r^3} + \frac{x^2y^2}{r^3}\right) + \frac{\partial \tilde{u}}{\partial \varphi}\left(\frac{2yx^3}{r^4} - \frac{2xy}{r^2} + \frac{4xy^3}{r^4} - \frac{2xy^3}{r^4}\right) + \frac{\partial^2 \tilde{u}}{\partial r^2}\left(\frac{x^4}{r^2} + \frac{2x^2y^2}{r^2} + \frac{y^4}{r^2}\right) +$$

$$+ \frac{\partial^2 \tilde{u}}{\partial r \partial \varphi}\left(-\frac{x^3y}{r^3} - \frac{x^3y}{r^3} + \frac{2x^3y}{r^3} - \frac{2xy^3}{r^3} + \frac{xy^3}{r^3} + \frac{xy^3}{r^3}\right) + \frac{\partial u}{\partial \varphi^2}\left(\frac{x^2y^2}{r^4} - \frac{2x^2y^2}{r^4} + \frac{x^2y^2}{r^4}\right) =$$

$$= r^2\frac{\partial^2 \tilde{u}}{\partial r^2}$$

Необх. усл Если f дифф в т. x^0 и x^0 экстр., то $\nabla f(x^0) = \vec{0}$

Дост. усл Если f дважды диф в т. x^0 и $d^2 f$ > 0, то x^0 – min; < 0, то x^0 – max; знакоопр. не дост. экстр.

$\nabla f(x^0) = \vec{0}$

§ 5 №18(2)

$$2x^2 + 2y^2 + u^2 + 8yu - u + 8 = 0$$

$u(x,y) = ?$

$$u'_x = -\frac{F'_x}{F'_u} = \frac{-4x}{2u + 8y - 1} \qquad 2u + 8y \neq 1$$

$$u'_y = -\frac{F'_y}{F'_u} = \frac{-4y + 8u}{2u + 8y - 1}$$

$u_x = 0 \Rightarrow x = 0$

$y = -2u$

$2 \cdot 4u^2 + u^2 + 8u(-2u) - u + 8 = 0$

$-7u^2 - u + 8 = 0$

$$\begin{cases} u_1 = 1 \\ u_2 = -\frac{8}{7} \end{cases}$$

$(x;y)=(0;\frac{16}{7})$ $u=-\frac{8}{7}$

$(x;y)=(0;-2)$ $u=1$

$$u''_{xx}=-\frac{u(2u+8y-1)-u_x(2u'_x)}{(2u+8y-1)^2}$$

$$u''_{yx}=\frac{u_x(2u'_y+8)}{(2u+8y-1)^2}$$

$$u''_{yy}=-\frac{(u+8u'_y)(2u+8y-1)-(u+8u)(2u'_y+8)}{(2u+8y-1)^2}$$

$$u''_{xx}\Big|_{(0;-2)}=\frac{-4(2-16-1)-0}{(1-16-1)^2}=\frac{60}{16^2}$$

$$u''_{xy}\Big|_{(0;-2)}=0$$

$$u''_{yy}=\frac{-(u+0)(2-16-1)-(-8+8)(\ )}{16^2}=\frac{60}{16^2}$$

$$d^2u\Big|_{(0;-2)}=\frac{60}{16^2}dx^2+\frac{60}{16}dy^2 \leftarrow \text{Это } \min$$

26.09

Условный экстремум

$\exists$ на области $G\subset\mathbb{R}^n$ заданы $f(x);\varphi_1(x)\dots\varphi_m(x)$ $m<n$

$\exists$ мн-во $E=\{x\in\mathbb{R}^n \mid \varphi_1(x)=\dots=\varphi_m(x)=0\}$

Опр Точка $x^0\in G$ наз (строгого) условным лок. ф-ции f при условиях $\varphi_1(x)=\dots=\varphi_m(x)=0$, если $\exists\delta>0:\forall x\in\mathring{U}_\delta(x_0)\cap E \to f(x)\ (<)\leqslant f(x^0)$

Необх усл условного экстр

$\exists$ $f(x),\varphi_1(x);\dots;\varphi_m(x)$ нек диф в $U(x^0)$ и ещё знач $\operatorname{rank}\left(\frac{\partial\varphi_i}{\partial x_j}\right)(x^0)=m$

Если x_0 явл усл. экстр, то $\frac{\partial L}{\partial x_i}(x^0)=0$ $i=\overline{1;n}$ $\left(L=f(x)+\sum_{i=1}^{m}\lambda_i\varphi_i(x)\ ;\ \lambda_i\in\mathbb{R}\right)$ ф-я Лагранжа

$\varphi_i(x^0)=0$ $i=\overline{1;m}$

Дост усл усл. экстр

$\exists$ $f;\varphi_1\dots\varphi_m$ дважды нек диф в $U(x^0)$ и выт. необх условия. Если ещё кв-ая

$$d\varphi_i(x^0) = \sum_{j=1}^{m} \frac{\partial \varphi_i}{\partial x_j}(x^0)\, dx_j = 0$$

$i = \overline{1,m}$ кв. ф. d^2L зад:

а) „+" ому $\Rightarrow$ x^0 - строг усл min

б) „−" ому $\Rightarrow$ x^0 - строг усл max

в) не знакоому $\Rightarrow$ x^0 не усл экстр

§5 №21(2)

$u = 1 - 4x - 8y \qquad x^2 - 8y^2 = 8$

$L = 1 - 4x - 8y + \lambda(x^2 - 8y^2 - 8)$

необх усл

$$\frac{\partial L}{\partial x} = -4 + 2\lambda x = 0$$

$$x^2 - 8y^2 - 8 = 0$$

$$\frac{\partial L}{\partial y} = -8 - 16\lambda y = 0$$

$\lambda \neq 0 \qquad x = 2/\lambda \qquad y = -\frac{1}{2\lambda}$

$$\frac{4}{\lambda^2} - 8\frac{1}{4\lambda^2} = 8$$

$$\frac{2}{\lambda^2} = 8 \qquad \lambda = \pm\frac{1}{2}$$

1) $\lambda = \frac{1}{2}$ $(4; -1)$

2) $\lambda = -\frac{1}{2}$ $(-4; 1)$ ← КАНДИДАТЫ

Выборы прошли, кандидаты остались

© Кузнецов Я. А.

Дост усл

$$\frac{\partial^2 L}{\partial x^2} = 2\lambda \qquad \frac{\partial^2 L}{\partial x \partial y} = 0 \qquad \frac{\partial^2 L}{\partial y^2} = -16\lambda$$

$$d^2L = (dx\ dy) \begin{pmatrix} 2\lambda & 0 \\ 0 & -16\lambda \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = 2\lambda\, dx^2 - 16\lambda\, dy^2 = 2\lambda(dx^2 - 8dy^2) =$$

$d\varphi = 0$

$\varphi = x^2 - 8y - 8$

$2x\,dx - 16y\,dy = 0$

$x\,dx - 8y\,dy = 0$

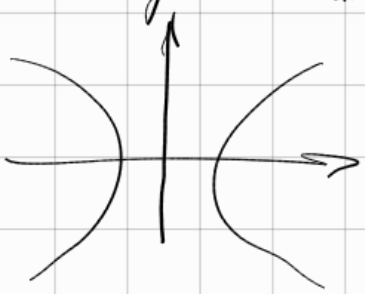
$dx = 8\frac{y}{x}dy$ $\quad = 2\lambda\left(64\frac{y^2}{x^2} - 8\right)dy^2$

$$d^2L\big|_{(1)} = 2\tfrac{1}{2}\left(64\cdot\tfrac{1}{16} - 8\right)dy^2 = -4\,dy^2 < 0 \Rightarrow (4;1) - \text{т. max} \qquad u = -7$$

$$d^2L\big|_{(2)} = 2\cdot\left(\tfrac{1}{2}\right)\left(64\cdot\tfrac{1}{16} - 8\right)dy^2 = 4\,dy^2 > 0 \Rightarrow (-4;1) - \text{т. min} \qquad u = 9$$

Второй способ

$x^2 - 8y = 8 \qquad u = 1 - 4x - 8y$



1) $x = 2\sqrt{2}\,\mathrm{ch}\,t$, $y = \mathrm{sh}\,t$, $t \in \mathbb{R}$

$u = 1 - 8\sqrt{2}\,\mathrm{ch}\,t - 8\,\mathrm{sh}\,t$

$u'_t = 0$ ... экстремум ф-ии одной переменной

2) $x = -2\sqrt{2}\,\mathrm{ch}\,t$, $y = \mathrm{sh}\,t$, $t \in \mathbb{R}$

$u = 1 + 8\sqrt{2}\,\mathrm{ch}\,t - \mathrm{sh}\,t$

$u' = \ldots 0$

§5 №26 (3)

$u = xy + yz \qquad x^2 + y^2 = 0$

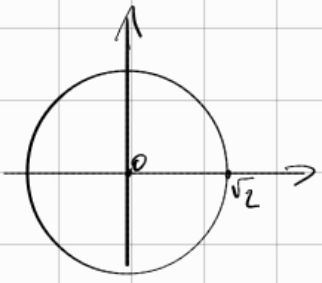
$y + z = 2$

$y \geq 0$

$z = 2 - y$

$u = x(2 - y) + 2y - y^2 \qquad x^2 + y^2 = 2$

$y \geq 0$



$x = \sqrt{2}\cos\varphi$

$y = \sqrt{2}\sin\varphi$

$\varphi \in (0;\pi)$

$u = 2\cos\varphi\sin\varphi + 2\sqrt{2}\sin\varphi - 2\sin^2\varphi$

$u'_\varphi = 2\cos 2\varphi + 2\sqrt{2}\cos\varphi - 4\sin\varphi\cos\varphi = 0$

$2\cos^2\varphi - 1 + \sqrt{2}\cos\varphi - 2\sin\varphi\cos\varphi = 0$

$\sin\varphi = \dfrac{2\cos^2\varphi + \sqrt{2}\cos\varphi - 1}{2\cos\varphi}$

$$\frac{4\cos^4\varphi + 2\cos^2\varphi + 1 - 4\cos^2\varphi - 2\sqrt{2}\cos\varphi + 4\sqrt{2}\cos^3\varphi}{4\cos^2\varphi} + \cos^2\varphi = 1$$

$8\cos^4\varphi - 6\cos^2\varphi + 4\sqrt{2}\cos^3\varphi - 2\sqrt{2}\cos\varphi + 1 = 0$

Вообще-то экстремальная (Матрица Guessa)

Вообще-то Дифференциально

Фурри ???

$8t^4 + 4\sqrt{2}t^3 - 6t^2 - 2\sqrt{2}t + 1 = 0$

$z = t\sqrt{2} \qquad t = z/\sqrt{2}$

$2z^4 + 2z^3 - 3z^2 - 2z + 1 = 0$

$(z^2 - 1)(2z^2 + 2z - 1) = 0$

$$\begin{cases} z = \pm 1 \\ z = \dfrac{-1 \pm \sqrt{3}}{2} \end{cases}$$

$\begin{cases} t = \pm \frac{1}{\sqrt{2}} \\ t = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases}$ $\begin{cases} \cos\varphi = \pm \frac{1}{\sqrt{2}} \\ \cos\varphi = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}} \end{cases}$

(1) $\begin{cases} x = 1 \\ y = 1 \end{cases}$ (2) $\begin{cases} x = -1 \\ y = 1 \end{cases}$ (3) $\begin{cases} x = \frac{-1-\sqrt{3}}{2} \\ y = \frac{\sqrt{3}-1}{2} \end{cases}$ (4) $\begin{cases} x = \frac{-1+\sqrt{3}}{2} \\ y = \dots \end{cases}$

$u''_{\varphi\varphi} = -4\sin\varphi\cos\varphi + 4\cos\varphi(-\sin\varphi) + 2\sqrt{2}(-\sin\varphi) - 4\cos\varphi\cos\varphi - 4\sin\varphi(-\sin\varphi) =$

$= -4\sin 2\varphi - 2\sqrt{2}\sin\varphi - 4\cos 2\varphi$

$\varphi = \frac{\pi}{4}$

$u''_{\varphi\varphi}\Big|_{\frac{\pi}{4}} = -4 - 2 - 0 = -6 < 0 \Rightarrow (1;1)$ т. max

§5 № 31(3)

$u = x + y + z \qquad x^2 + y^2 \le z \le 1$

Найти наиб и наим знач

1) Проверим экстремумы внутри

$\nabla u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow$ нет экстр внутри области

1. "Крышечка" $z = 1 \quad x^2 + y^2 \le 1$

$u = x + y + 1 \qquad x^2 + y^2 \le 1$



проверим внутри круга

$\nabla u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

нет экстр

на границе $u = x + y + 1 \quad x^2 + y^2 = 1$

$x = \cos\varphi \quad y = \sin\varphi$

$u = \cos\varphi + \sin\varphi + 1$

$u' = -\sin\varphi + \cos\varphi = 0$

$\sin\varphi = \cos\varphi \qquad \varphi = \frac{\pi}{4} + \pi k$

$\begin{cases} x = \frac{1}{\sqrt{2}} \\ y = \frac{1}{\sqrt{2}} \end{cases} \quad u = 1 + \sqrt{2}$

$\begin{cases} x = -\frac{1}{\sqrt{2}} \\ y = -\frac{1}{\sqrt{2}} \end{cases} \quad u = 1 - \sqrt{2}$

2). Тарелочка

$x^2+y^2=z$ $\quad u=x+y+z$

$x^2+y^2\le 1$

$x^2+y^2=z$, $x^2+y^2\le 1$

$x^2+y^2=z$, $x^2+y^2=1$ ← граница куда

$u=x+y+x^2+y^2$

$x^2+y^2<1$

$\nabla u=\begin{pmatrix}1+2x\\1+2y\end{pmatrix}=0$ $\quad \begin{cases}x=-\frac{1}{2}\\ y=-\frac{1}{2}\end{cases}$

$\eta=1+\sqrt{2}$

$m=-\frac{1}{2}$

$u=-\frac{1}{2}-\frac{1}{2}+\frac{1}{4}+\frac{1}{4}=-\frac{1}{2}$

03.09

Кратные интегралы

I Определение

$\exists X$ - изм. мн-во на $\mathbb{R}^n$

$\tau(X)=\{X_j\}_{j=1}^{\nu}: \mu(X_j\cap X_k)=0 \quad j\neq k$

$X=\bigcup\limits_{j=1}^{\nu} X_j$

Опр: $\operatorname{diam} Y=\sup\limits_{y_1,y_2\in Y}\rho(y_1;y_2) \quad Y\in\mathbb{R}^n$

Опр $|\tau|=\max\limits_{j=1\dots\nu}\operatorname{diam} X_j$ - мелкость разбиения

В каждом X_j выберем ξ_j.

$\{\xi_j\}_{j=1}^{\nu}$ - набор точек

Опр. $\exists f$-ция на изм. мн-ве $X\subset\mathbb{R}^n$

Говорят, что f интегр на X по Риману,

если $\exists I\in\mathbb{R}\ \forall\varepsilon>0\ \exists\delta>0\ \forall\tau(X)\ |\tau|<\delta\ \forall\{\xi_j\}_{j=1}^{\nu}\ \mapsto |\sigma(f;\tau;\{\xi_j\})-I|<\varepsilon$

$\sigma=\sum\limits_{j=1}^{\nu} f(\xi_j)\mu(X_j)$ ← сумма Римана

I наз интегралом (кратным). Обозн $\int\limits_X f(x)\,dx=\int\dots\int\limits_X f(x)\,dx_1\dots dx_n$

II Критерий Коши

$\exists f$ оп на измеримом $X \subset \mathbb{R}^n$

f интегр $\Leftrightarrow \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall \tau' \ |\tau'| < \delta \ \forall \{\xi_j'\}_{j=1}^{n'} \ \forall \tau'' \ |\tau''| < \delta \ \forall \{\xi_j''\}_{j=1}^{n''} \hookrightarrow |\sigma(f;\tau';\{\xi'\}) - \sigma(f;\tau'';\{\xi''\})| < \varepsilon$

Теорема f интегр на X - огр. $\Rightarrow$ f оп на X

III Критерий Дарбу

$\exists f$ оп на $X \subset \mathbb{R}^n$ (измеримое)

$\bar{S}_\tau = \sum_{j=1}^{n} m_j \mu(X_j)$ $\quad m_j = \inf_{X_j} f(x)$ $\quad I^* = \inf_\tau \bar{S}_\tau$

$S_\tau = \sum_{j=1}^{n} M_j \mu(X_j)$ $\quad M_j = \sup_{X_j} f(x)$ $\quad I_* = \sup_\tau S_\tau$

Теорема $\exists f$ оп на изм. X, тогда

1) f интегр $\Leftrightarrow I^* = I_*$
2) $\forall \varepsilon > 0 \ \exists \tau : \bar{S}_\tau - S_\tau < \varepsilon$

IV Достаточное условие интегрируемости

$\exists f$ оп на замкн изм. $X \subset \mathbb{R}^n$

Мн-во точек разрыва f имеет меру 0 $\Rightarrow$ f интегр на X

---

§8 №23 указать ф-ию инт на изм. мн-ве но не интегр на этом мн-ве

$$f(x) = \frac{1}{x} \ ; x \in (0;1)$$

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§8 №36.1 указать измеримое X и оп на нем f такие что $|f|$ интегр f не интегр

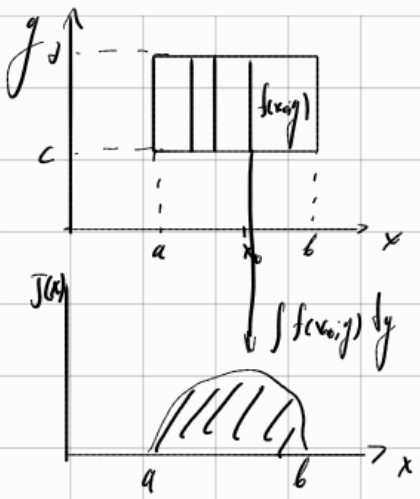
$$f(x) = \begin{cases} -1, & x \in [0;1] \cap \mathbb{Q} \\ 1, & x \in [0;1] \setminus \mathbb{Q} \end{cases}$$

Т6

$\exists f(x;y)$ определена на $[a;b] \times [c;d]$ $\quad a<b \ ; \ c<d$

а) Верно ли, что если

$f(x_0;y)$ интегр на $[c;d]$ $\forall x_0$ и $J(x) = \int_a^b f(x;y)\,dy$ интегр, то кратный интеграл $\exists \int_a^b \int_c^d f(x;y)\,dx\,dy$



$$\int_a^b \left(\int_c^d f(x;y)\,dy \right) dx$$ повторный интеграл

Пем! $[a;b]=[0;1]$ $[c;d]=\left[\frac{\pi}{2};\frac{3\pi}{2}\right]$

$$f(x;y)=\begin{cases} \sin y, & x\in[0;1]\cap\mathbb{Q} \\ 0, & x\in[0;1]\setminus\mathbb{Q} \end{cases}$$

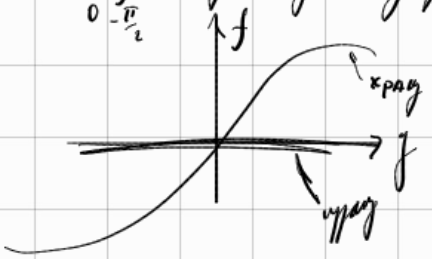
$$\int_c^d f(x;y)\,dy = \int_{-\pi/2}^{\pi/2} f(x;y)\,dy$$

$x\in\mathbb{Q}$: $\int_{-\pi/2}^{\pi/2} \sin y\,dy = 0$

$x\notin\mathbb{Q}$: $\int_{-\pi/2}^{\pi/2} 0\,dy = 0$

$$\int_a^b \left(\int_c^d f(x;y)\,dy \right) dx = \int_0^b 0\,dx = 0$$

$\int_0^1 \int_{-\pi/2}^{\pi/2} f(x;y)\,dx\,dy$ не сущ?



$I^* > 0$

$I_* < 0$

А по поле? Дарбу

б) Верно ли, что если f инт на $[a;b]\times[c;d]$, то $f(x_0;y)$ инт на $[c;d]$ $\forall x_0\in[a;b]$

Пем $[a;b]=[0;1]$ $[c;d]=[0;1]$

$$f(x)=\begin{cases} 0, & x\neq\frac{1}{2} \\ 1, & x=\frac{1}{2}\ y\in[0;1]\cap\mathbb{Q} \\ 0, & x=\frac{1}{2}\ y\in[0;1]\setminus\mathbb{Q} \end{cases}$$

мера мн-ва точек разрыва $=0$ $\int_a^b\int_c^d f(x;y)\,dx\,dy\in\mathbb{R}$

$\int_0^1 f(x_0;y)\,dy$ $x_0=\frac{1}{2}$ не сущ

Неделим

Опр Мн-во $X\subset\mathbb{R}^2$ наз элементарной отн Оу, если $X=\{(x;y)\in\mathbb{R}^2 \mid a\le x\le b;\ \varphi(x)\le y\le\psi(x)\}$, где $\varphi(x)$ и $\psi(x)$ нпр на $[a;b]$

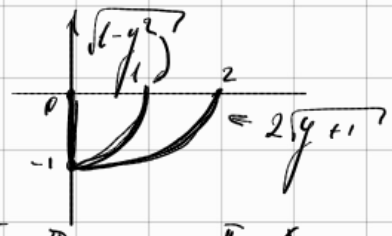


Теорема] f нпр на элементарной отн Оу мн-ве X. Тогда $\iint_X f(x;y)\,dx\,dy = \int_a^b dx \int_{\varphi(x)}^{\psi(x)} f(x;y)\,dy$

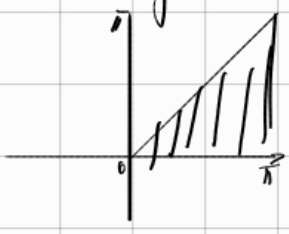
$\psi(x) = \begin{cases} \sqrt{1-x^2} & x \in [0;1] \\ 0 & x \in (1;2] \end{cases}$

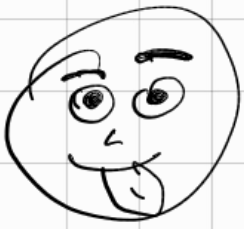
§ 8 №83(6)

$$\int_{-1}^{0} dy \int_{\sqrt{1-y^2}}^{2\sqrt{y+1}} f(x;y)\,dx = \int_{0}^{2} dx \int_{\frac{x^2}{4}-1}^{\psi(x)} f(x;y)\,dy$$



§ 8 №85(2) $\int_0^{\pi} dy \int_y^{\pi} \frac{\sin x}{x}\,dx = \int_0^{\pi} dx \int_0^{x} \frac{\sin x}{x}\,dy = \int_0^{\pi} dx \frac{\sin x}{x}\, y\Big|_0^x = \int_0^{\pi} \sin x\,dx = 2$

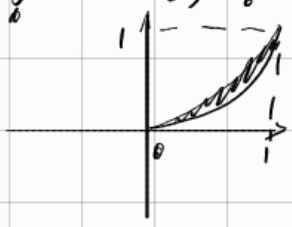




10.10 №90(2;6)

2) $\iint_G \frac{y}{x^2}\,dx\,dy = \int_0^1 dx \int_{x^3}^{x^2} \frac{y}{x^2}\,dy = \int_0^1 dx \left(\frac{x^2}{2} - \frac{x^4}{2}\right) = \frac{1}{6} - \frac{1}{10} = \frac{1}{15}$

$G = \{0 < x;\ x^3 \le y \le x^2\}$



Машина одеет Standoff 2 !!!

6) $\iint_G (x+2y)\,dx\,dy = \int_2^3 dx \int_x^{2x} (x+2y)\,dy = \int_2^3 dx\,(xy+y^2)\Big|_x^{2x} = \int_2^3 4x^2\,dx = \frac{4}{3}x^3\Big|_2^3 = \frac{76}{3}$

G ог $\begin{matrix} y = x \\ x \to 2 \end{matrix}$ $\begin{matrix} y = 2x \\ x = 3 \end{matrix}$

Замена переменных в кратном интеграле

Теорема $\exists X \subset \mathbb{R}^n$ $Y \subset \mathbb{R}^n$, $X; Y$ - измеримо

φ - отображение, переводящее $\overline{Y}$ на $\overline{X}$.

пусть: 1) φ - взаимно однозначно

2) φ - непр диф

Если $f(x)$ интегр на X, тогда $\int_X f(x)\,dx = \int f(\varphi(y))\,|\det J|\,dy$

$J = \frac{\partial \varphi}{\partial x} = \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1} & \cdots & \frac{\partial \varphi_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial \varphi_n}{\partial x_n} & \cdots & \frac{\partial \varphi_n}{\partial x_n} \end{pmatrix}$

№31(6) $\iint\limits_G (2y-x)\,dx\,dy$ $\quad G = \{y(y-x)\le 2;\ x(x+y)\le 3\}$

I

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} -y & 2y-x \\ 2x+y & x \end{pmatrix}$$

$$|J| = -xy - (2y-x)(2x+y) = -xy - (4xy + 2y^2 - 2x^2 - xy) = -4xy + 2x^2 - 2y^2$$

$u = -x$ $\quad x = -u$

$v = -y$ $\quad y = -v$

$$\mathcal{J} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

$|\det \mathcal{J}| = 1.$ $\quad \tilde{G} = \{v(v-u)\le 2,\ u(u+v)\le 3\}$

$$I = \iint\limits_{\tilde{G}} (u-2v)\cdot 1\,du\,dv =$$

DeMurist

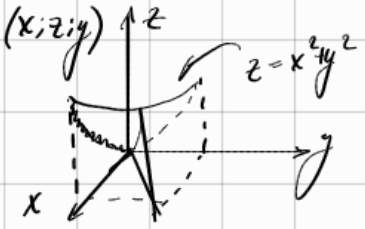
$$= -\iint\limits_{\tilde{G}} (2v-u)\,du\,dv = -I$$

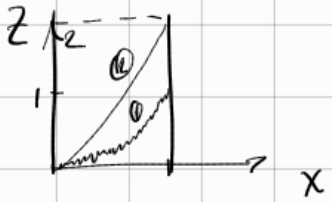
$\Rightarrow I = -I = 0$

№ 132(5)

$$\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f(x;y;z)\,dz = \left[\int_0^1 dx \int_0^{x^2} dz \int_0^1 f\,dy + \int_0^1 dx \int_{x^2}^{2x^2} dz \int_{\sqrt{z-x^2}}^1 f\,dy\right]$$

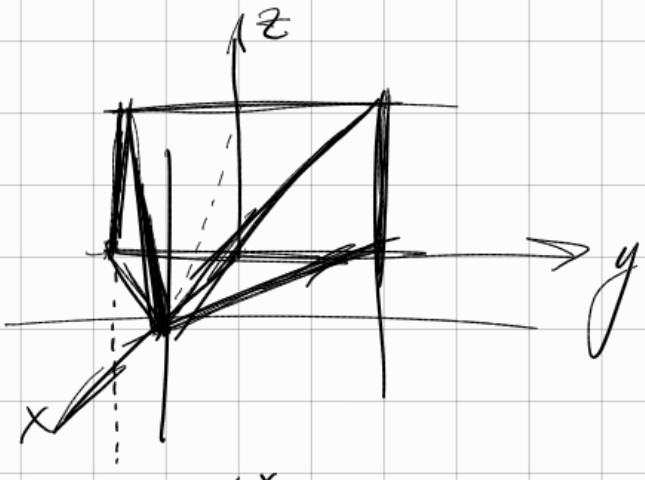
a) (x;z;y)





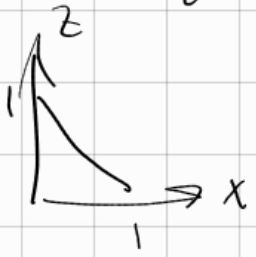
№ 139(3)

$$\iiint_G (x+z)\,dx\,dy\,dz \quad (=)$$



$x+y=1$
$x-y=1$
$x+z=1$
$z=0$
$x=0$

$$(=) 2\int_0^1 dx \int_0^{1-x} dz \int_0^{1-x} (x+z)\,dy = 2\int_0^1 dx \int_0^{1-x} (x+z)(1-x)\,dz =$$





$$= 2\int_0^1 dx \int_0^{1-x} x(1-x) + z(1-x)\,dz =$$

$$y = 2\int_0^1 dx \left(x(1-x)^2 + \frac{1}{2}(1-x)^3\right) =$$

$$= 2\int_0^1 (1-x)\left(x + \frac{1}{2} - \frac{x}{2}\right) dx =$$

$$= \int_0^1 (1-x)^2 (x+1)\,dx = \int_0^1 (1-x)(1-x^2)\,dx =$$

$$= \frac{5}{12}$$

№ 172 (1)

$$\int_0^a dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-1}} f(x_n)\,dx_n = \frac{1}{(n-1)!}\int_0^a f(t)(a-t)^n\,dt \text{, где } [0;+\infty)$$

По индукции: $\int_0^a f(t)\,dt = \frac{1}{0!}\int_0^a f(t)(a-t)^{1-1}\,dt$ – верно

Предположение $\int_0^a dx_1 \dots \int_0^{x_{k-1}} f(x_k)\,dx_k = \frac{1}{(k-1)!}\int_0^a f(t)(a-t)^{k-1}\,dt$

$\int_0^a dx_1 \dots \int_0^{x_{k-1}} dx_k \underbrace{\int_0^{x_k} f(x_{k+1})\,dx_{k+1}}_{F(x_k)}$

$\int_0^a dx_1 \dots \int_0^{x_{k-1}} F(x_k)\,dx_k = \frac{1}{(k-1)!}\int_0^a F(t)(a-t)\,dt = \frac{1}{(k-1)!}\int_0^a \int_0^t f(y)\,dy\,(a-t)^{k-1}\,dt =$

$= \frac{1}{(k-1)!}\left[-\frac{1}{k}(a-t)^k \int_0^t f(y)\,dy \Big|_0^a + \frac{1}{k}\int_0^a f(t)(a-t)^k\,dt\right] =$

$= \frac{1}{k!}\int_0^a f(t)(a-t)^k\,dt$

$\iint_{G^*} f(\rho\cos\varphi;\rho\sin\varphi)\rho\,d\rho\,d\varphi$

§8 №100(4)

$\iint_G f(x;y)\,dx\,dy =$ (=)

Полярные координаты

$x = \rho\cos\varphi$

$y = \rho\sin\varphi$

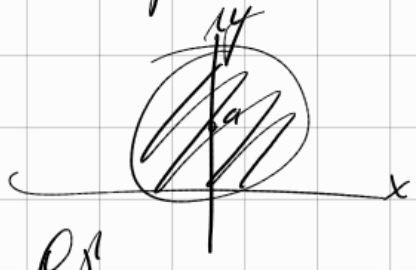
$dx\,dy = |\det J|\,d\rho\,d\varphi = \rho\,d\rho\,d\varphi$

$\det J = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \varphi} \end{vmatrix} = \begin{vmatrix} \cos\varphi & -\rho\sin\varphi \\ \sin\varphi & \rho\cos\varphi \end{vmatrix} = \rho$

$G: x^2+y^2 \le 2ay \quad a>0$

$x^2+y^2-2ay+a^2 \le a^2$

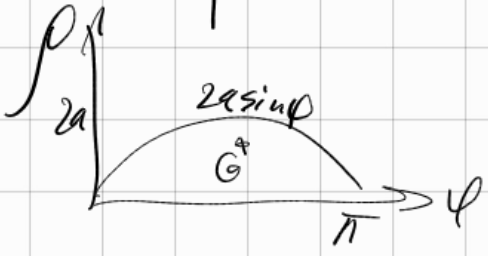
$x^2+(y-a)^2 \le a^2$



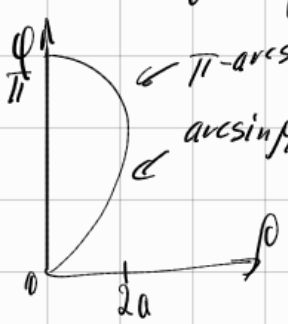
$x^2+y^2=2ay$

$\rho^2 = 2a\rho\sin\varphi$

$\rho = 2a\sin\varphi$



(=) $\int_0^{\pi} d\varphi \int_0^{2a\sin\varphi} f(\rho\cos\varphi;\rho\sin\varphi)\rho\,d\rho$



(=) $\int_0^{2a} \rho\,d\rho \int_{\arcsin\rho/2a}^{\pi-\arcsin\rho/2a} f(\rho\cos\varphi;\rho\sin\varphi)\,d\varphi$

№107 (2)

Вычислить

$$\iint\limits_G y\,dx\,dy, \quad G=\{x^2+y^2\le 2x,\ x\ge y\}$$

$(x-1)^2+y^2\le 1$



$x'=x-1$

$y'=y$

$$\iint y'\,dx'dy' \quad G^*=\{x'^2+y'^2\le 1,\ x'+1\ge y'\}$$



$$I_2=\int_0^1\int_{-\pi}^{\pi/2}\rho\sin\varphi\cdot\rho\,d\rho\,d\varphi=\int_0^1\rho^2d\rho\int_{-\pi}^{\pi/2}\sin\varphi\,d\varphi=$$

$$=\frac{1}{3}\cdot(-1)=-\frac{1}{3}$$

$$I_1=\int_{-1}^{0}\int_{0}^{1+x'}y'\,dx'dy'=\int_{-1}^{0}dx'\int_0^{1+x'}y'dy'=\int_{-1}^{0}\frac{(y')^2}{2}\Big|_0^{1+x'}dx'=\int_{-1}^{0}\frac{(1+x')^2}{2}dx'=\frac{1}{6}(1+x)^3\Big|_{-1}^{0}=\frac{1}{6}$$

$I=I_1+I_2$

Ответ: $-\frac{1}{6}$

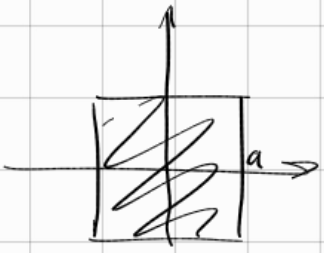
§8 №110

1) $\iint\limits_G e^{-(x^2+y^2)}dx\,dy=$ $\quad G=\{x^2+y^2\le a^2,\ x\ge 0,\ y\ge 0\}$

$$=\int_0^{\pi/2}d\varphi\int_0^a e^{-(\rho^2\cos^2\varphi+\rho^2\sin^2\varphi)}\rho\,d\rho=\int_0^{\pi/2}d\varphi\int_0^a e^{-\rho^2}\rho\,d\rho=\frac{\pi}{2}\cdot\frac{1}{2}e^{-\rho^2}\Big|_a^0=\frac{\pi}{4}(1-e^{-a^2})$$

2) (!) $\frac{\sqrt{\pi}}{2}\sqrt{1-e^{-a}}<\underbrace{\int_0^a e^{-x^2}dx}_{I}<\frac{\sqrt{\pi}}{2}\sqrt{1-e^{-2a^2}}$

$$I^2=\int_0^a e^{-x^2}dx\int_0^a e^{-y^2}dy=\iint_0^a e^{-(x^2+y^2)}dx\,dy$$



$$\iint\limits_{x^2+y^2\le a^2}e^{-(x^2+y^2)}dx\,dy<I^2<\iint\limits_{x^2+y^2\le 2a^2}e^{-(x^2+y^2)}dx\,dy$$

$$\sqrt{\frac{\pi}{4}(1-e^{-a^2})}<I<\sqrt{\frac{\pi}{4}(1-e^{-2a^2})}$$ ЧТД

3) $\frac{\sqrt{\pi}}{2}\sqrt{1-e^{-a^2}} < \int_0^a e^{-x^2}dx < \frac{\sqrt{\pi}}{2}\sqrt{1-e^{-2a^2}}$

$a \to +\infty$ $\quad$ $a \to +\infty$ $\quad$ $a \to +\infty$

$\pi/2$ $\quad$ $\frac{\pi}{2}$ $\quad$ $\pi/2$

$$\int_0^{+\infty} e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$$

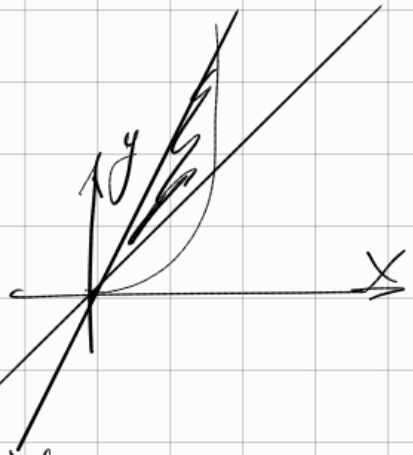
№124(3)

$$\iint_G e^{x^4/y^2}dxdy \quad \textcircled{=}$$

$y = 2x$

$y = x$

$y = x^2$



$$\textcircled{=} \int_{\pi/4}^{\arctg 2} d\varphi \int_0^{\frac{\sin\varphi}{\cos^2\varphi}} e^{\frac{\rho^4\cos^4\varphi}{\rho^2\sin^2\varphi}} \rho\, d\rho \quad \boxed{=} \quad \begin{matrix} y = x^2 \\ \rho\sin\varphi = \rho^2\cos\varphi \\ \rho = \frac{\sin\varphi}{\cos\varphi} \end{matrix}$$

$$\boxed{=} \frac{1}{2}\int_{\pi/4}^{\arctg 2} \frac{\sin^2\varphi}{\cos^4\varphi} d\varphi \int_0^{\frac{\sin\varphi}{\cos^2\varphi}} e^{\frac{\rho^2\cos^4\varphi}{\sin^2\varphi}} 2\rho\, \frac{\cos^4\varphi}{\sin^2\varphi}\, d\rho$$

$$= \frac{1}{2}\int_{\pi/4}^{\arctg 2} \frac{\sin^2\varphi}{\cos^2\varphi} e^{\frac{\cos^2\varphi}{\sin^2\varphi}\rho^2}\Big|_0^{\frac{\sin\varphi}{\cos^2\varphi}} = \frac{1}{2}\int_{\pi/4}^{\arctg 2} \frac{\sin^2\varphi}{\cos^2\varphi}(e-1)\,d\varphi = \frac{1}{2}\int_{\pi/4}^{\arctg 2} tg^2\varphi(e-1)\,dtg\varphi$$

$$= \frac{1}{2}\cdot\frac{1}{3}(e-1)\,tg^3\varphi\Big|_{\pi/4}^{\arctg 2} = \frac{1}{6}(e-1)(8-1) = \frac{7}{6}(e-1)$$